Lab 5 - List of improving moves

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Description of the problem

Each instance has n = 200 nodes with coordinates (x, y) and a cost. Select exactly **half of the nodes** (k = 100) and connect them into a **closed cycle** that minimises **cycle length + sum of node costs**. Distances are Euclidean, rounded to integers.

Algorithms

Steepest local search with lab 3's best neighborhood (**inter-route node exchange + edges exchange**) from lab 3's **random tours** (seed 42 + i), reusing the improving moves of one evaluation from a **list of improving moves (LM)**.

Intuition

Problem: steepest LS evaluates all 14 950 moves again after every move, although one move changes only a few edges, so almost every result is the same as one step before.

Idea: keep the improving moves found in a list, best first, and apply them while they still fit the tour; the whole neighborhood is evaluated again only when none fits: on TSPA 9 instead of 134 times per run. The tours are as good as with the full search.

Core calculations

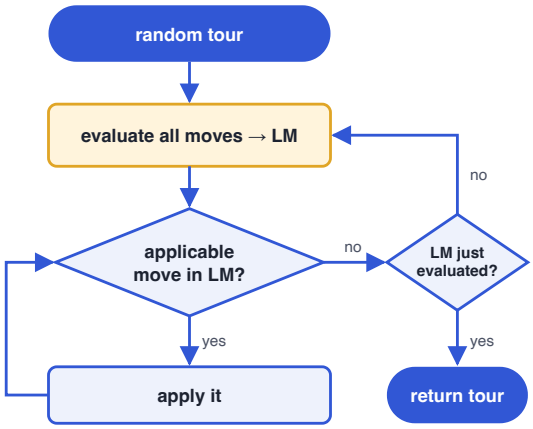
For every move we calculate **Δ = the change of the objective** (Δ < 0: improving). **LM** keeps the improving moves, best first, stored by the **nodes of the removed edges**, not by positions.

Node exchange, selected v (between p and s) → unselected u:
`Δ = c(u) − c(v) + d(p,u) + d(u,s) − d(p,v) − d(v,s)`

Edges exchange, remove a→b, c→d, both relative directions:
`Δ = d(a,c) + d(b,d) − d(a,b) − d(c,d)` same
`Δ = d(a,d) + d(b,c) − d(a,b) − d(c,d)` inverted

Δ uses only these nodes, so a **stored Δ stays exact** while its removed edges are in the tour. **Browsing LM** from the best:

Removed edges in the tour	Action
at least one missing	remove from LM
relatively reversed	keep, browse further
same relative direction	apply, remove from LM



Complexity

Method	Complexity per step	Δ per run
Steepest LS, edges exchange, random start	O(nk)	2 003 300
Steepest LS, edges exchange, candidate moves, random start	O(k), lists O(n² log n)	227 305
Steepest LS, edges exchange, list of improving moves, random start	browse ≤ O(nk), rebuild O(nk log nk)	177 201

Δ per run is the TSPA average over 200 runs.

Explanation

- **Full steepest search:** every step computes the Δ of every move, k(n – k) inter-route + k(k – 1)/2 intra-route ≤ nk → **O(nk)** per step: `k(n – k) + k(k – 1)/2 = 14 950`; 134 steps per run.
- **Candidate moves:** a step computes Δ only for moves adding a candidate edge, 2 moves for each of the 10 candidates of each of the k tour nodes → **O(k)** per step: at most `2 × 10 × k = 2 000` (about 1 800 on average); 126 steps per run. Once per run the 10 candidates of every node are found by sorting → **lists O(n² log n)**.
- **LM:** a step computes no Δ; it browses LM from the best, O(1) per stored move and at most all of them → **browse ≤ O(nk)**, then applies the move in O(k). Only when no stored move is applicable, a rebuild computes the Δ of both directions of every edge pair, `k(n – k) + k(k – 3) = 19 700`, and sorts the improving ones → **rebuild O(nk log nk)**; 9 rebuilds for 154 moves per run.
- So **LM computes 11× fewer Δ than the full search** and slightly fewer than candidate moves, but it also pays for sorting and browsing LM.

Pseudocode

Simplified from `python/solver.py`, the alternative algorithm of the assignment. The assignment's "add improving moves to LM" is done by rebuilding LM: the moves left in LM are found again by the full evaluation with the same Δ, so the result is the same without duplicates. The search stops when a freshly evaluated LM has no applicable move: the tour is then a local optimum of the full neighborhood, as in lab 3.

```
def solve(seed):
    tour = random_start(rng(seed))              # same start as lab 3
    return local_search(tour)

def local_search(tour):
    while True:
        moves = evaluate_all_moves(tour)        # LM
        move = take_applicable_move(moves, tour)
        if move is None:
            return tour                         # local optimum
        while move is not None:                 # reuse LM without evaluating again
            apply_move(tour, move)
            move = take_applicable_move(moves, tour)

def evaluate_all_moves(tour):
    moves = []
    for every v in tour (between p and s), every unselected u:
        add (Δ, EXCHANGE_SELECTED, p, v, s, u) if Δ < 0
    for every pair of non-adjacent tour edges a -> b, c -> d:
        add (Δ, SWAP_EDGES, a, b, c, d) if Δ < 0     # same relative direction
        add (Δ, SWAP_EDGES, a, b, d, c) if Δ < 0     # relatively inverted
    return moves sorted from the best

def take_applicable_move(moves, tour):
    for move in moves, from the best:
        status = move_status(tour, move)
        if status == OBSOLETE:   remove move from moves
        if status == APPLICABLE: remove move from moves; return move
        # NOT_APPLICABLE_NOW: keep it, browse further
    return None

def move_status(tour, move):                    # edge_direction: +1 x -> y, -1 y -> x, 0 no edge
    if move is (EXCHANGE_SELECTED, p, v, s, u):
        if edge_direction(p, v) == 0 or edge_direction(v, s) == 0 or u in tour:
            return OBSOLETE
        return APPLICABLE                       # p, v, s consecutive: same direction
    if move is (SWAP_EDGES, a, b, c, d):
        first, second = edge_direction(a, b), edge_direction(c, d)
        if first == 0 or second == 0: return OBSOLETE
        if first != second:           return NOT_APPLICABLE_NOW
        return APPLICABLE

def apply_move(tour, move):
    if EXCHANGE_SELECTED: u takes the position of v
    if SWAP_EDGES:        i, j = positions of a and c (or of b and d if both edges are reversed)
                          reverse tour[i+1 .. j]
```

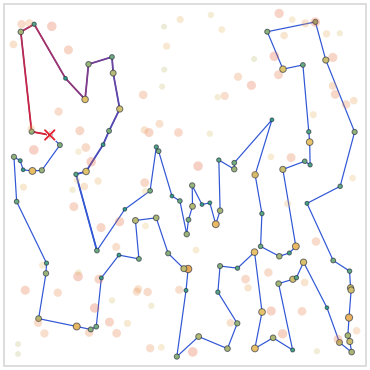
Implementation note. LM is stored worst first, so the best move is at the end of the array: the ~11 400 removals per run then shift only a few kept moves, not the whole list (best-first storage made C, C++ and Rust as slow as Python).

Results

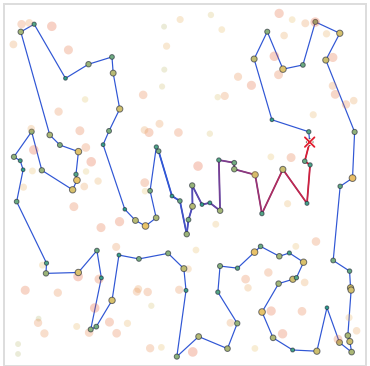
Objective = cycle length + node costs, 200 runs per method. Cells: av (min – max), best of each per instance in bold.
All best solutions were checked with the solution checker: **yes**.
Results are identical in all implemented languages: **yes**.

Method	TSPA	TSPB
Steepest LS, edges exchange, random start	76225 (74248 – 79388)	**52461** (50265 – **55173**)
Steepest LS, edges exchange, greedy start	**72997** (**71955** – **74682**)	57019 (52566 – 62301)
Steepest LS, edges exchange, candidate moves, random start	80062 (76007 – 85627)	52560 (**49902** – 55299)
Steepest LS, edges exchange, list of improving moves, random start	76119 (73975 – 79089)	52493 (50369 – 55548)

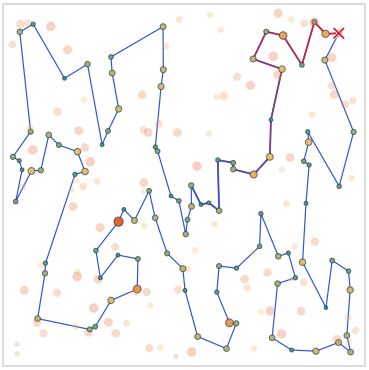
Visualisation (Instance TSPA)



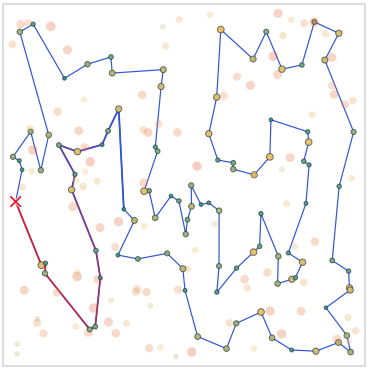
Steepest LS, edges exchange, random start
74248 (length 22259 + cost 51989)



Steepest LS, edges exchange, greedy start
71955 (length 21123 + cost 50832)

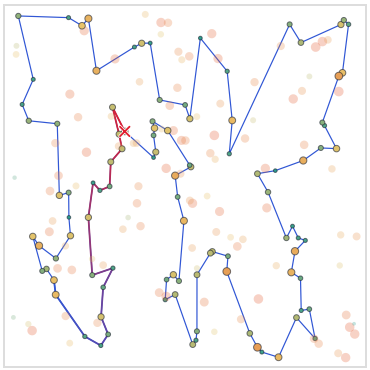


Steepest LS, edges exchange, candidate moves, random start
76007 (length 22323 + cost 53684)

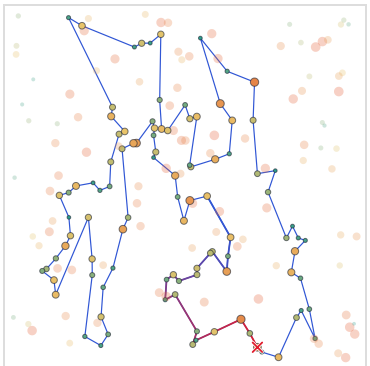


Steepest LS, edges exchange, list of improving moves, random start
73975 (length 22573 + cost 51402)

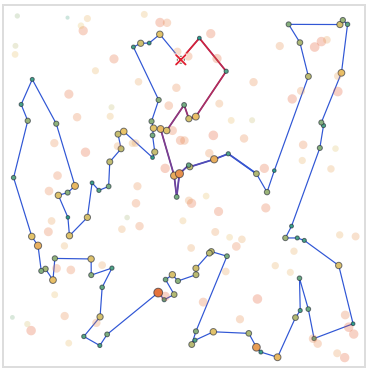
Visualisation (Instance TSPB)



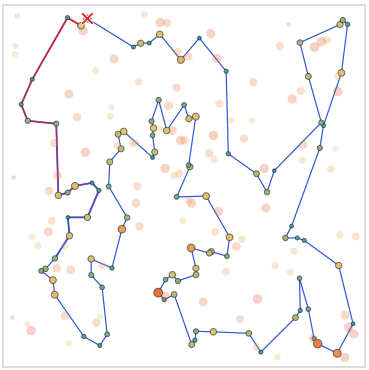
Steepest LS, edges exchange, random start
50265 (length 19741 + cost 30524)



Steepest LS, edges exchange, greedy start
52566 (length 17522 + cost 35044)



Steepest LS, edges exchange, candidate moves, random start
49902 (length 19786 + cost 30116)



Steepest LS, edges exchange, list of improving moves, random start
50369 (length 19937 + cost 30432)

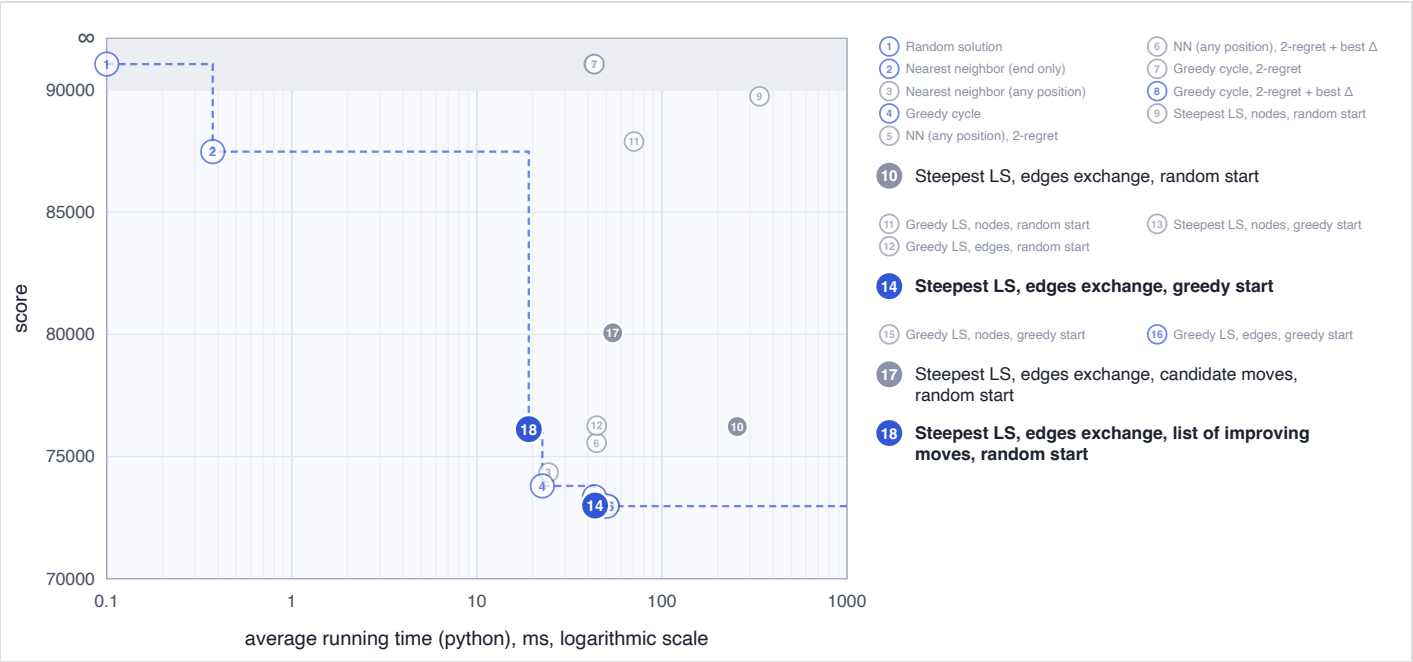
Running time

Average time of one run **in ms** (algorithm only, same runs as above); the fastest language per row is in bold.

Instance TSPA

Method	Python	C	C++	Rust
Steepest LS, edges exchange, random start	255.24	3.48	**3.38**	4.13
Steepest LS, edges exchange, greedy start	43.54	0.89	**0.87**	0.95
Steepest LS, edges exchange, candidate moves, random start	54.18	**0.68**	1.24	1.03
Steepest LS, edges exchange, list of improving moves, random start	19.08	1.10	0.96	**0.92**

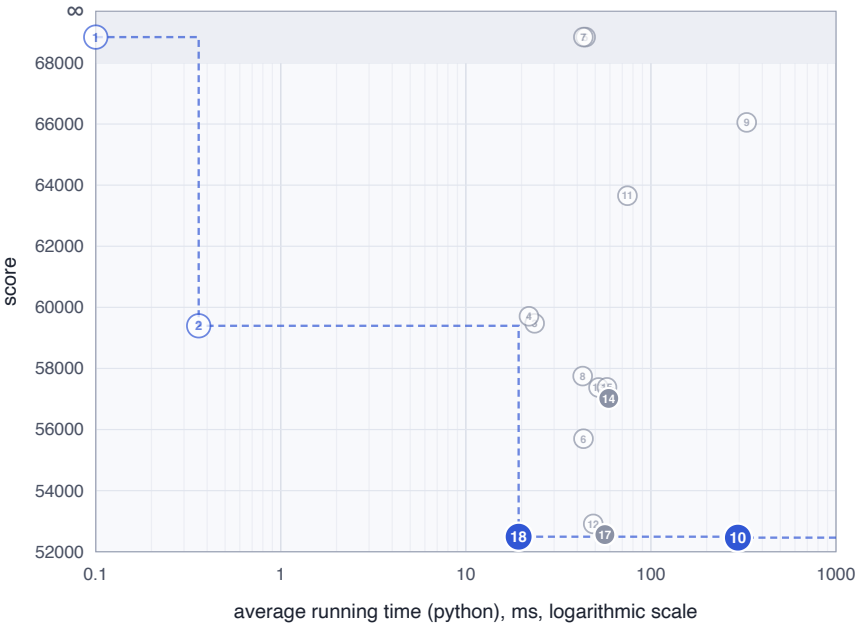
Pareto plot (Instance TSPA)



Instance TSPB

Method	Python	C	C++	Rust
Steepest LS, edges exchange, random start	294.92	3.54	**3.44**	4.28
Steepest LS, edges exchange, greedy start	59.17	1.12	**1.04**	1.16
Steepest LS, edges exchange, candidate moves, random start	56.34	**0.66**	1.17	0.98
Steepest LS, edges exchange, list of improving moves, random start	19.28	0.94	0.81	**0.77**

Pareto plot (Instance TSPB)



Best solutions (0-based node indices)

Instance TSPA

Steepest LS, edges exchange, list of improving moves, random start
168, 95, 97, 127, 171, 2, 187, 30, 160, 123, 152, 100, 132, 76, 4, 180, 28, 141, 87, 45, 73, 135, 181, 192, 158, 138, 182, 57, 64, 88, 78, 153, 107, 170, 146, 159, 130, 47, 147, 101, 55, 29, 178, 184, 136, 186, 154, 62, 143, 33, 9, 54, 8, 142, 164, 1, 169, 3, 50, 93, 106, 113, 89, 42, 151, 150, 174, 176, 72, 134, 58, 84, 131, 36, 177, 52, 179, 53, 140, 144, 67, 185, 35, 156, 16, 51, 139, 112, 71, 157, 91, 80, 48, 31, 108, 49, 60, 116, 149, 63

Instance TSPB

Steepest LS, edges exchange, list of improving moves, random start
42, 147, 116, 149, 28, 145, 158, 177, 125, 26, 154, 140, 97, 170, 115, 6, 29, 132, 197, 36, 175, 134, 45, 60, 71, 155, 40, 33, 64, 47, 38, 21, 30, 183, 69, 160, 172, 136, 133, 120, 117, 72, 46, 86, 184, 3, 151, 27, 66, 67, 24, 34, 25, 152, 181, 167, 73, 59, 107, 85, 51, 82, 87, 179, 88, 99, 171, 74, 20, 153, 14, 2, 9, 93, 110, 118, 90, 129, 80, 127, 10, 114, 61, 105, 94, 81, 70, 187, 196, 54, 192, 162, 76, 57, 89, 156, 48, 96, 198, 121

Source code

labs/lab5 (Python, C, C++, Rust)

Conclusions

- **LM reaches the quality of the full steepest search** (76 119 vs 76 225 on TSPA, 52 493 vs 52 461 on TSPB). Both end in a local optimum of the same neighborhood; only the order of the applied moves differs, because moves created after the last evaluation wait for the next one.
- Evaluating the neighborhood 9 instead of 134 times per run makes the search **3.2–5.6× faster** in C, C++ and Rust and **13–15× faster** in Python (19 ms vs 255–295 ms). In Python almost all the time is spent on evaluations, while compiled code also pays for sorting and browsing LM.
- Compared with candidate moves (lab 4), LM is **about as fast** (C 1.10 vs 0.68 ms, Rust 0.92 vs 1.03 ms on TSPA) **without their quality loss** on TSPA (76 119 vs 80 062).
- LM speeds the search up but **does not move the best results so far**: on TSPA greedy-start LS from lab 3 is still best on average (72 971), and the best TSPB solution still comes from candidate moves (49 902 vs 50 369 here).